

High-speed silicon modulator with band equalization

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Electro-optic modulation up to 70 Gbit/s has been demonstrated using a silicon Mach-Zehnder modulator with a bias voltage of -1.5 V. In a wide frequency range from DC, an increasing input impedance of the modulator was designed to equalize its electro-optic frequency response. Without a bias voltage, the 3 dB bandwidth was measured as 35 GHz and it is predicted to be as high as 55 GHz at -3 V bias. Frequency responses of the modulator operated with counter-propagating waves were tested to verify the proposed prediction model. © 2014 Optical Society of America

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The broadband electro-optical (E/O) modulator is a critical component for encoding light by either intensity or phase modulation. To develop a low-cost and on-chip integrated optical links, various kinds of silicon-based modulators have been proposed and exhibited dramatic performance improvements in recent years [1–10]. Broad bandwidths in excess of 65 GHz for hybrid-silicon electro-absorption modulators [3] and plasmonic modulators [4] have been demonstrated. Because of the fabrication simplicity, the all-silicon modulators formed monolithically in silicon are preferred. Among them, the silicon depletion-mode Mach-Zehnder modulator (MZM) can offer high modulation speed, broadband operation spectral, and high thermal tolerance. It has played an important role in some integrated optical links [1,10]. However, because of the relatively long phase shifter, the 3 dB bandwidth of these reported broadband silicon depletion-mode MZMs was limited, even though high reverse bias voltages were applied to decrease the depletion capacitance of these MZMs. It has been mostly demonstrated that -4 V $\sim$ -6 V reverse bias voltages were needed for high-speed electro-optical modulations up to 50–60 Gbit/s [5–8]. Moreover, with the increase of reverse bias voltage, the increasing slope of depletion width in the PN junction declines; the modulation efficiency of a silicon depletion-mode MZM decreases as well. To keep pace with the demands for high-speed optical interconnects, increasing the bandwidth of the silicon depletion-mode MZM with a low bias voltage is urgently needed.

In previous research, it has been found that a low termination resistor can cause an initial rise in the frequency response of a modulator [11], which may expand its bandwidth [12] and benefit the high-speed E/O modulation [13] as with a high-pass equalizer filter [14]. Meanwhile, this phenomenon is related to the cost of modulating power available to drive the modulator. In this Letter, a broadband silicon depletion-mode MZM is demonstrated and characterized. Over a broad frequency range from DC, increasing input impedances were designed to achieve flat E/O modulation responses for the MZM at different bias voltages. Without a bias voltage, the 3 dB E/O bandwidth of the proposed silicon MZM was measured as 35 GHz, and it is predicted to be as high as 55 GHz at -3 V bias. An open and clear eye

diagram at 70 Gbit/s with an extinction ratio (ER) of 6.1 dB has been recorded with a low bias voltage of -1.5 V.

As the driving signal propagates through the phase shifter of a depletion-mode MZM, it gets significantly attenuated at higher frequencies. However, there is less effect of attenuation at lower frequencies. This preferential attenuation of high frequency components has the impact of increasing the rise and fall time of the driving signal, which degrades the MZM's performance in high-speed modulation. A flat E/O modulation response benefits the modulated optical signal's integrity for the MZM, which is crucial in some applications such as an optical analog communication system. Figure 1(a) shows the microscope image of the proposed MZM. It can be modeled as a transmission-line circuit as shown in Fig. 1(b). V_G is the amplitude of the driving signal and Z_G is the source impedance. The characteristic impedance Z_0 and propagation constant γ of the phase shifter can be evaluated by an equivalent circuit model proposed for the silicon depletion-mode MZM [15]. A titanium nitride (TiN) film was deposited in the end of the MZM as the terminal impedance Z_L . The input driving voltage V_{in} of the MZM is given by [16]

$$V_{in} = V_G \times \frac{Z_{in}}{Z_G + Z_{in}}, \quad (1)$$

where

$$Z_{in} = Z_0 \times \frac{Z_L + Z_0 \times \tanh(\gamma \times l)}{Z_0 + Z_L \times \tanh(\gamma \times l)} \quad (2)$$

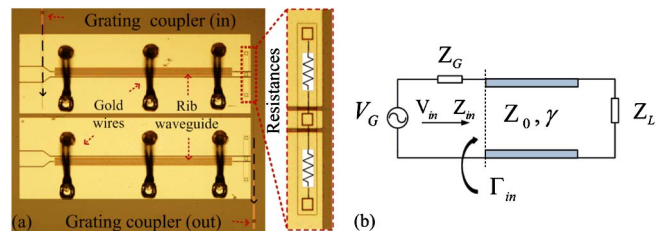


Fig. 1. (a) Microscope image of the MZM. (b) Equivalent circuit.

and

$$\gamma = \alpha + j\beta = a + j \frac{\omega \times n_e}{c}. \quad (3)$$

In Eqs. (1)–(3), Z_{in} is the input impedance, α is the attenuation constant, β is the phase constant, l is the length of the phase shifter, n_e is the effective index of the driving signal and its electrical wavelength λ_e can be expressed as $\lambda_e = 2\pi/\beta$. c is the speed of light in the free space and $\omega = 2\pi f$ is the angular frequency. As shown in Eq. (1), a continuously rising Z_{in} with frequency can elevate V_{in} to compensate for the E/O modulation response of a modulator in high frequency. However, as quantified by Eq. (2), Z_{in} is frequency dependent because of the wave nature of the driving signal. Therefore, the terminal resistor, length, and transmission-line parameters (Z_0 and γ) of the phase shifter should be designed specifically for the band equalization over a broad frequency range.

The doping profile and substrate of the phase shifter are similar to that described in Ref. [15], but a 10 nm wide intrinsic region was left between each interleaved PN junction. A 1 μm thick aluminum CPW electrode was designed with the signal line width of 20 μm and gap width of 5 μm . The MZM was fabricated in a CMOS-compatible fab at the Institute of Microelectronics [17]. Gold wire bridges were utilized to suppress the parasitic coupled-slotline mode excited by the MZM. The junction capacitance and resistance were evaluated as [15] $C_J = 274/\sqrt{0.92 - V_{Bias}} (fF/mm)$ and $R_J = 0.011 (\Omega \cdot m)$. When the bias voltage (V_{Bias}) varies between 0 and -5 V, Fig. 2(a) shows the calculated Z_0 and n_e of the phase shifter in a wide frequency range. The phase shifter was carefully chosen as 750 μm and TiN resistances were designed to be 23 Ω , which is slightly less than the designed Z_0 of the phase shifter. Figures 2(b) and 2(c) show the calculated input impedances and E/O modulation responses of the MZM. At -5 V bias, the variations of Z_0 and n_e are negligible with the change of frequency. The real part of Z_{in} can increase monotonically up to the frequency of 36 GHz, for which the phase shifter is approximately equal to $\lambda_e/4$ of the driving signal. Meanwhile, if no bias voltage is applied, Z_0 rises from 25 to 27 Ω and n_e falls from 3.6 to 3.3 when the frequency increases from DC to 40 GHz. This trend expands the maximum frequency to be 42 GHz, up to which the Z_{in} can rise continuously from DC, even through the n_e is higher than that at -5 V bias. The E/O modulation responses of the designed MZM were calculated by Eq. (10) in Ref. [15]. As shown in Fig. 1(c), over a broad frequency range from 0 to 40 GHz, the swing amplitudes of these E/O modulation responses are no more than 1 dB when the bias voltage varies between -1 and -5 V. Even without a reverse bias, the 3 dB E/O modulation bandwidth of the MZM is over 40 GHz.

To characterize the frequency response performances of the MZM, the electrical reflection coefficient (Γ_{in}) of the MZM was measured at different bias voltages. Figure 3(a) shows the comparison of the Γ_{in} measured by a vector network analyzer and that calculated by the proposed circuit model, in which the fitting junction capacitance is $C_J = 260/\sqrt{1 - V_{Bias}} (fF/mm)$ and resistance in series is $R_J = 0.023 (\Omega \cdot m)$. It is indicated that

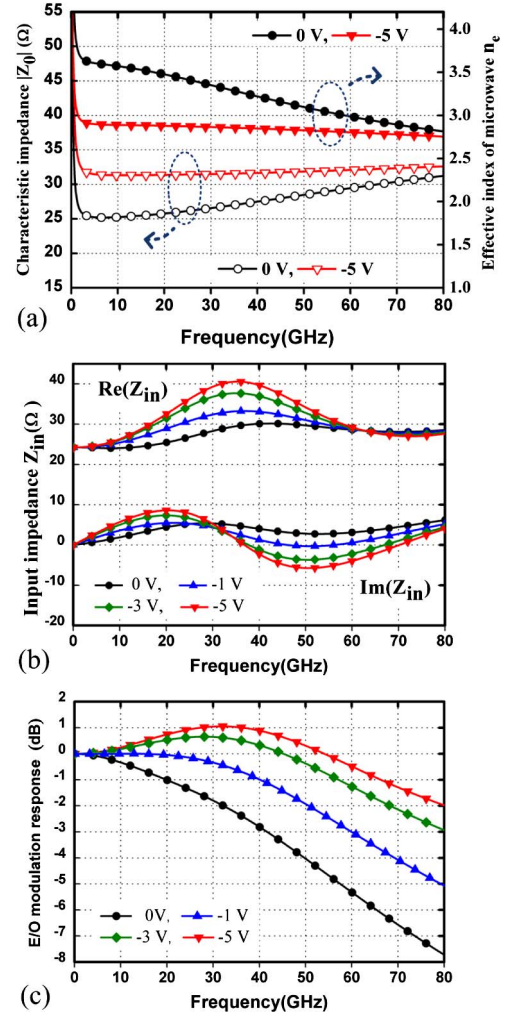


Fig. 2. Calculated properties of the MZM with different bias voltages: (a) characteristic impedance Z_0 and effective index of microwave n_e , (b) input impedance Z_{in} , and (c) E/O modulation response.

the 10 nm wide intrinsic region left in interleaved PN junctions decreased the junction capacitance and increased the resistance in series. Because of the skin effect, the TiN resistor was measured as 18 Ω at DC and increased to 18.8 Ω at 40 GHz. The parasitic inductance in series is approximately 16 pH. The input impedance Z_{in} of the MZM can be calculated by [16] $Z_{in} = Z_G(1 + \Gamma_{in})/(1 - \Gamma_{in})$. As shown in Fig. 3(b), with the increase of Z_{in} , E/O modulation responses of the MZM can be compensated up to about 35 GHz when the bias varying between 0 and 3 V. Figure 4(a) shows the measured E/O modulation responses of the MZM with the input optical wavelength set at the quadrature point. The E/O bandwidth of a modulator is defined as the frequency for which the optical intensity modulation depth has fallen to 70.7% of a reference level, causing a 3 dB reduction in the received signal [18]. If the reference level is chosen as the maximum modulation response, the 3 dB E/O bandwidth of the MZM was measured as 35 GHz at 0 V bias and is predicted as high as 55 GHz at -3 V bias. To verify the proposed prediction model, the E/O modulation responses of the modulator operated with counter-propagating waves were tested under the

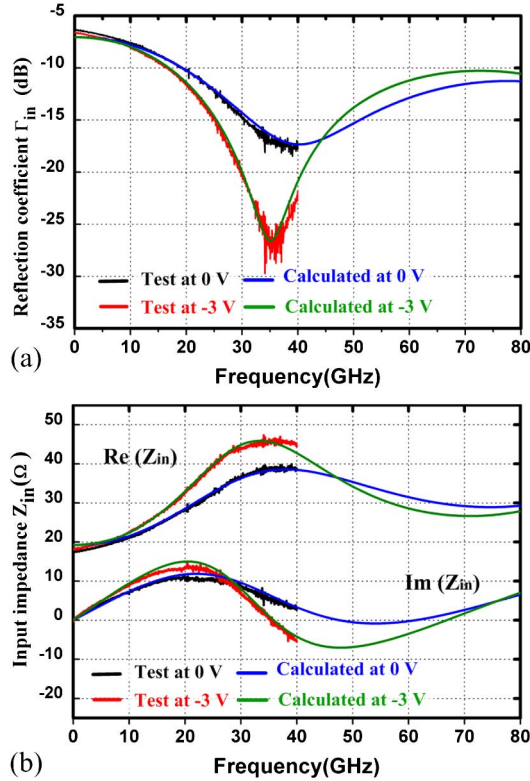


Fig. 3. Measured and calculated (a) reflection coefficient and (b) input impedance.

same condition. When the optical and electrical waves are traveling in opposite directions in the MZM, the resulting modulation response is limited by the velocity

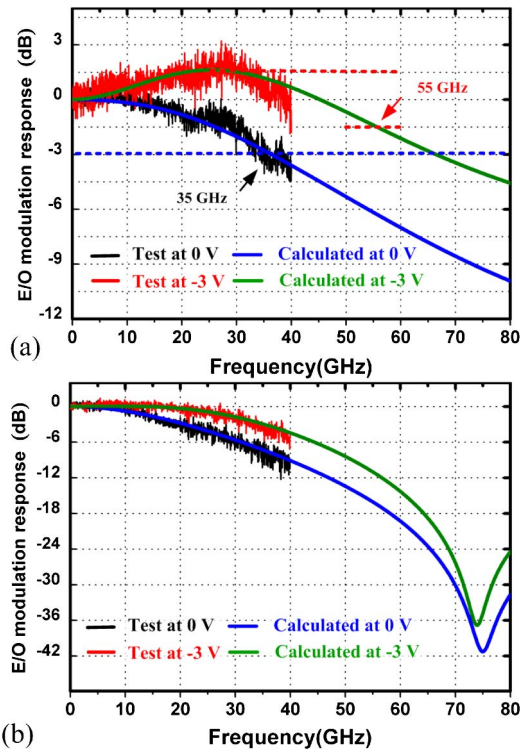


Fig. 4. E/O modulation responses of the MZM operated when optical and modulating waves were propagating: (a) in the same direction and (b) in opposite directions.

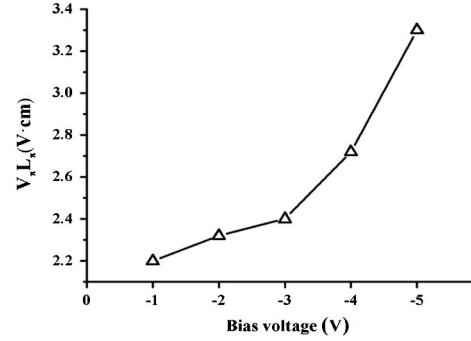


Fig. 5. Modulation efficiencies ($V_{\pi}L_{\pi}$) at different bias voltages.

mismatch between optical wave and the driving electrical signal. Mathematically, it can be calculated by replacing the n_o to be $-n_o$ [19]. Figure 4(b) compares the measured and calculated E/O modulation responses; in this case, good agreement was achieved as well at both 0 and -3 V bias.

Using the transfer function of an asymmetric MZM to fit the measured optical spectrum [20], the $V_{\pi}L_{\pi}$ of the phase shifter can be measured. As shown in Fig. 5, the $V_{\pi}L_{\pi}$ increased from 2.2 to 2.4 V·cm when V_{bias} was raised from -1 to -3 V. In addition, the decreasing slope of the modulation efficiency decreased rapidly when the V_{bias} was further raised to -5 V. The optical loss of the phase shifter was measured as 2.3 dB, and the insertion loss induced by light splitter and combiner were measured as 1.04 dB.

Eye diagrams of the modulated optical signals were tested at 60 Gbit/s and 70 Gbit/s to verify the high-speed modulation capability of the modulator. Differential electrical signals were amplified to drive the MZM on its two phase shifters. Figures 6(a) and 6(b) show the recorded driving signals with 30 dB attenuation. The amplitudes of these driving signals were measured as 5.75 V at 60 Gbit/s and 5.32 V at 70 Gbit/s. However, the input driving voltage V_{in} available for the MZM degraded because of the reflection caused by the impedance mismatch. Using a spectrum analysis method [21] and Eq. (1) as the transfer function, the amplitudes of V_{in} could be estimated as 3.5 V at 60 Gbit/s and 3.2 V at 70 Gbit/s. As shown in Figs. 6(c) and 6(d), clear and open optical eye diagrams were recorded at both 60 Gbit/s and 70 Gbit/s

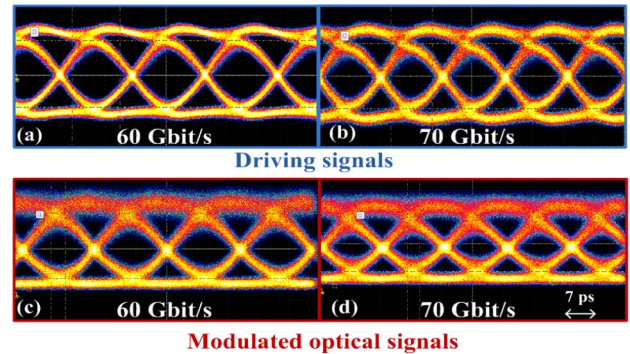


Fig. 6. Eye diagrams for driving signals with 30 dB attenuation at (a) 60 Gbit/s and (b) 70 Gbit/s. Modulated optical signals at (c) 60 Gbit/s, ER = 7.4 dB and (d) 70 Gbit/s, ER = 6.1 dB.

at a low bias voltage of -1.5 V. High dynamic extinction ratios (ER) were measured as 7.4 dB at 60 Gbit/s and 6.1 dB at 70 Gbit/s, even without considering the deterioration induced by the gain saturation of EDFA.

In conclusion, high-speed silicon MZM is experimentally demonstrated with a low bias voltage. With the band equalization design in a wide frequency range, flat modulation responses of the silicon MZM were obtained at different bias voltages. The method proposed in this Letter could be applied to achieve band equalization of the high-speed modulation of a modulator as well as an optical link, which may improve the channel capacity for high-speed optical interconnects in the future.

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References

1. M. Paniccia, Nat. Photonics **4**, 498 (2010).
2. G. T. Reed, G. Mashanovich, F. Y. Gardes, and D. J. Thomson, Nat. Photonics **4**, 518 (2010).
3. Y. Tang, J. D. Peters, and J. E. Bowers, Opt. Express **20**, 11529 (2012).
4. A. Melikyan, L. Alloatti, A. Muslija, D. Hillerkuss, P. C. Schindler, J. Li, R. Palmer, D. Korn, S. Muehlbrandt, D. Van Thourhout, B. Chen, R. Dinu, M. Sommer, C. Koos, M. Kohl, W. Freude, and J. Leuthold, Nat. Photonics **8**, 229 (2014).
5. D. Thomson, F. Gardes, J. Fedeli, S. Zlatanovic, Y. Hu, B. Kuo, E. Myslivets, N. Alic, G. Mashanovich, and G. Reed, IEEE Photon. Technol. Lett. **24**, 234 (2012).
6. P. Dong, L. Chen, C. Xie, L. L. Buhl, and Y. K. Chen, Opt. Express **20**, 6163 (2012).
7. X. Xiao, H. Xu, X. Li, Y. Hu, Z. Li, T. Chu, Y. Yu, and J. Yu, Opt. Express **21**, 4116 (2013).
8. X. Tu, T. Liow, J. Song, X. Luo, Q. Fang, M. Yu, and G. Lo, Opt. Express **21**, 12776 (2013).
9. M. Streshinsky, R. Ding, Y. Liu, A. Novack, Y. Yang, Y. Ma, X. Tu, E. Chee, A. Lim, P. Lo, T. Baehr-Jones, and M. Hochberg, Opt. Express **21**, 30530 (2013).
10. P. Dong, C. Xie, L. Chen, L. L. Buhl, and Y.-K. Chen, "112-Gb/s monolithic PDM-QPSK modulator in silicon," presented at the European Conference on Optical Communication, Amsterdam, The Netherlands, 2012, paper Th.3.B.1.
11. K. Kubota, J. Noda, and O. Mikami, IEEE J. Quantum Electron. **16**, 754 (1980).
12. S. H. Lin and S. Y. Wang, Appl. Opt. **26**, 1696 (1987).
13. L. Liao, A. Liu, J. Basak, H. Nguyen, M. Paniccia, D. Rubin, Y. Chetrit, R. Cohen, and N. Izhaky, Electron. Lett. **43**, 1196 (2007).
14. S. Hall and H. Heck, in *Advanced Signal Integrity for High-Speed Digital Design* (Wiley, 2009), Chap. 12.
15. H. Xu, X. Li, X. Xiao, Z. Li, Y. Yu, and J. Yu, IEEE J. Sel. Top. Quantum Electron. **20**, 3400110 (2014).
16. R. Ludwig and P. Bretchko, *RF Circuit Design: Theory and Applications* (Prentice-Hall, 2000).
17. T. Liow, K. Ang, Q. Fang, J.-F. Song, Y. Xiong, M. Yu, G. Lo, and D. Kwong, IEEE J. Sel. Top. Quantum Electron. **16**, 307 (2010).
18. R. G. Walker, IEEE J. Quantum Electron. **27**, 654 (1991).
19. W. R. Leeb, A. L. Scholtz, and E. Bonek, IEEE J. Quantum Electron. **QE-18**, 14 (1982).
20. H. Xu, X. Li, X. Xiao, Z. Li, Y. Yu, and J. Yu, Chin. Phys. B **22**, 114212 (2013).
21. S. Kaneko, H. Itamoto, T. Miyahara, and T. Hatta, "Impact of package resonance on eye diagram in high-speed optical modules," presented at the European Conference on Optical Communication, Amsterdam, The Netherlands, 2001, paper WE.P.12.